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Conditional probability and Bayes: updating beliefs with evidence

A test finds 90% of sick people and wrongly flags 10% of healthy people. You test positive. How likely are you sick?

Mathematics for AI and Machine Learning | Session 15 of 18 | Probability and statistics

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Today in one view: what you will leave with

By the end of today you can



Read joint, marginal and conditional probabilities from a table of counts, test for independence, and say what conditional independence means

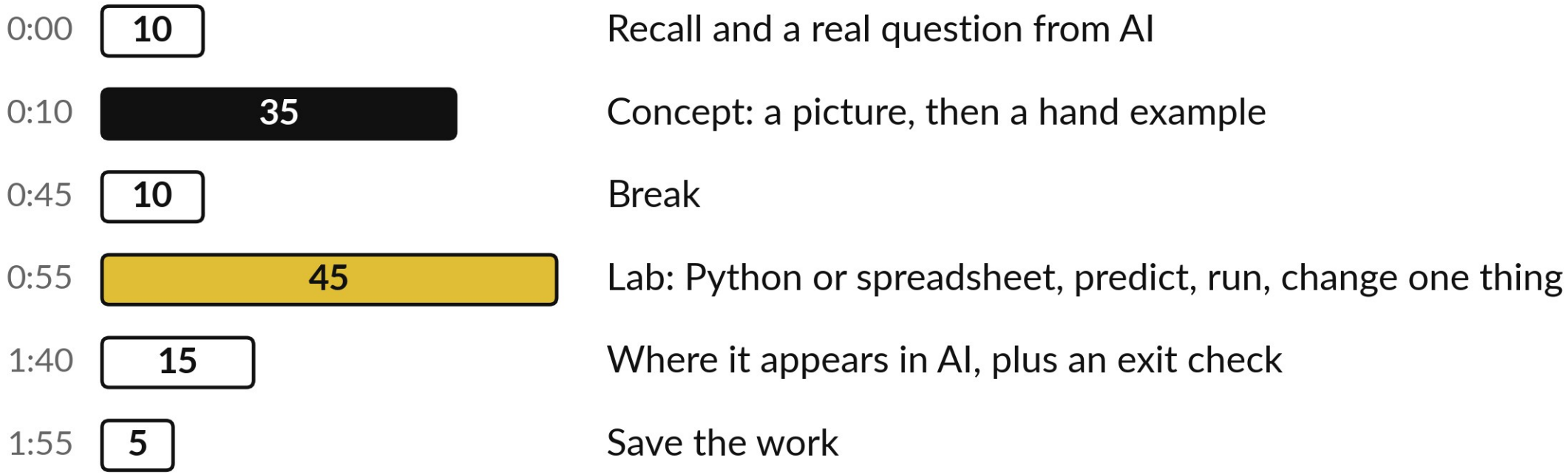


Apply Bayes theorem with natural frequencies, naming the prior, likelihood and posterior



Build a small Naive Bayes spam classifier by hand and in code, and read a model probability with care

Your two hours: counts, Bayes, a spam filter, then AI



Counts first, then Bayes with natural frequencies, then a spam filter by hand and in the lab.

Three questions from last time

1. Why can the same question get a different answer from a chat model?

2. What does a standard error measure?

3. An A/B test gives p = 0.039. What does that mean?

Answer aloud first. Then we check together.

Answers to the three questions from Session 14

Question	Short answer
Why can the same question get a different answer from a chat model?	The next word is drawn at random from its probabilities. A new random number u can land in a different slice.
What does a standard error measure?	How much a sample mean wanders from sample to sample: sigma divided by the square root of n.
An A/B test gives p = 0.039. What does that mean?	A gap this large would be rare if there were truly no difference. It is not a 3.9% chance that the feature does nothing.

If an answer surprised you, look back at Session 14 before the lab.

Words we use today

	Word	What it means
	**Conditional probability**	P(A | B): the probability of A once we know B happened.
	**Conditional independence**	Two words are independent within each class (spam, ham), though not overall. Independence itself is from Session 13.
	**Prior, likelihood and posterior**	The belief before evidence, the chance of the evidence if the hypothesis is true (for a test: its sensitivity), and the updated belief.
	**Bayes' theorem**	The rule P(H | E) = P(E | H) P(H) / P(E): the posterior from the likelihood, the prior and the evidence.
	**Naive Bayes classifier**	A classifier that multiplies per-word likelihoods, assuming the words are independent within each class.

A table of counts shows joint and marginal probabilities

100 emails	Contains "free"	No "free"	Total
Spam	**28**	12	**40**
Ham (not spam)	6	54	**60**
Total	**34**	**66**	**100**

Illustrative: 100 emails, 40 spam and 60 ham, sorted by whether they contain the word 'free'.

Joint probability: P(spam and free) = 28 / 100 = 0.28 (gold).

Marginal probability: one event on its own, from a row or column total. P(free) = 34 / 100 = 0.34.

P(spam) = 40 / 100 = 0.40.

A conditional probability looks only at the emails where the condition holds

100 emails	Contains "free"	No "free"	Total
Spam	**28**	12	**40**
Ham (not spam)	6	54	**60**
Total	**34**	**66**	**100**

Given that an email contains 'free', only the 34 emails in the first column matter.

P(spam | free) = 28 / 34 = 0.8235.

P(free | spam) = 28 / 40 = 0.70.

The two are different questions with different answers.

Joint, marginal and conditional probability

Conditional

$$P(A \mid B) = \frac{P(A \text{ and } B)}{P(B)}$$

Product rule

$$P(A \text{ and } B) = P(B)\,P(A \mid B)$$

Symbol	What it means	In our data
P(A and B)	The joint probability: both events happen.	P(spam and free) = 0.28
P(B)	The marginal probability of the condition.	P(free) = 0.34
P(A | B)	The conditional probability: A given B.	0.28 / 0.34 = 0.8235

The product rule is the same formula rearranged: 0.34 × 0.8235 = 0.28.

Recall independence from Session 13: 'free' and spam fail the test, so the word carries information

	A	B	C
		Contains "free": observed	Expected if independent
1	Spam	28	13.6
2	Ham	6	20.4

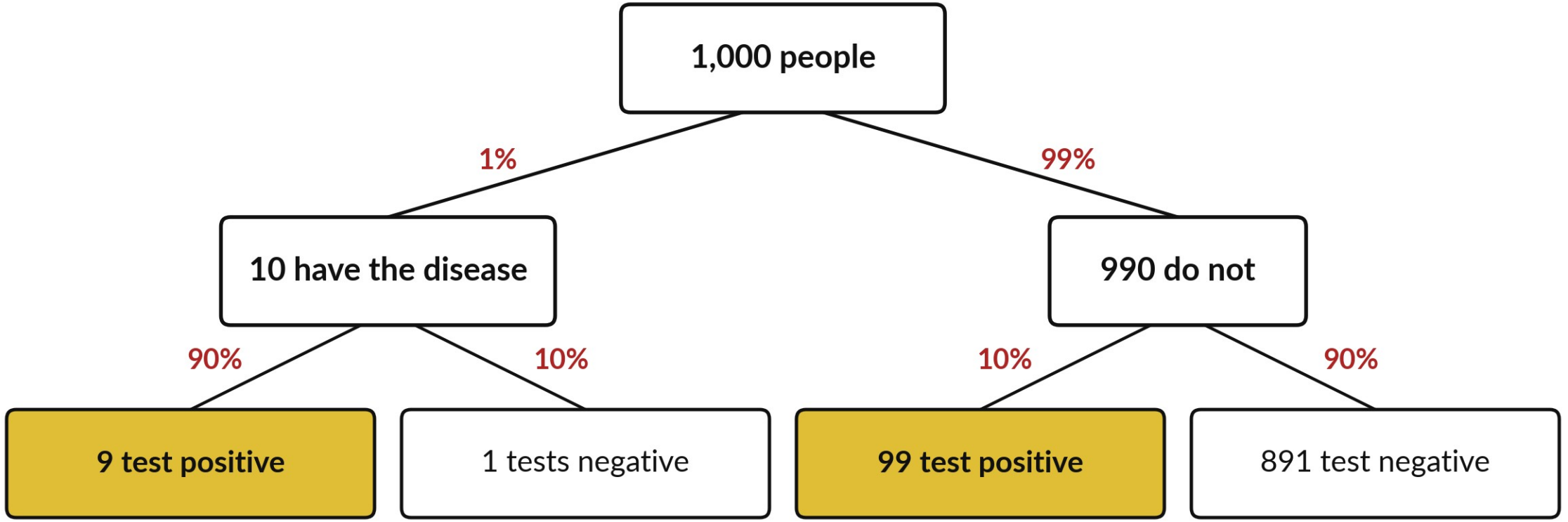
Independent (Session 13): P(A and B) = P(A) × P(B). Test it on the counts.

If independent: P(spam) × P(free) = 0.40 × 0.34 = 0.136, so 13.6 emails.

Observed: 28 emails. Far more than 13.6.

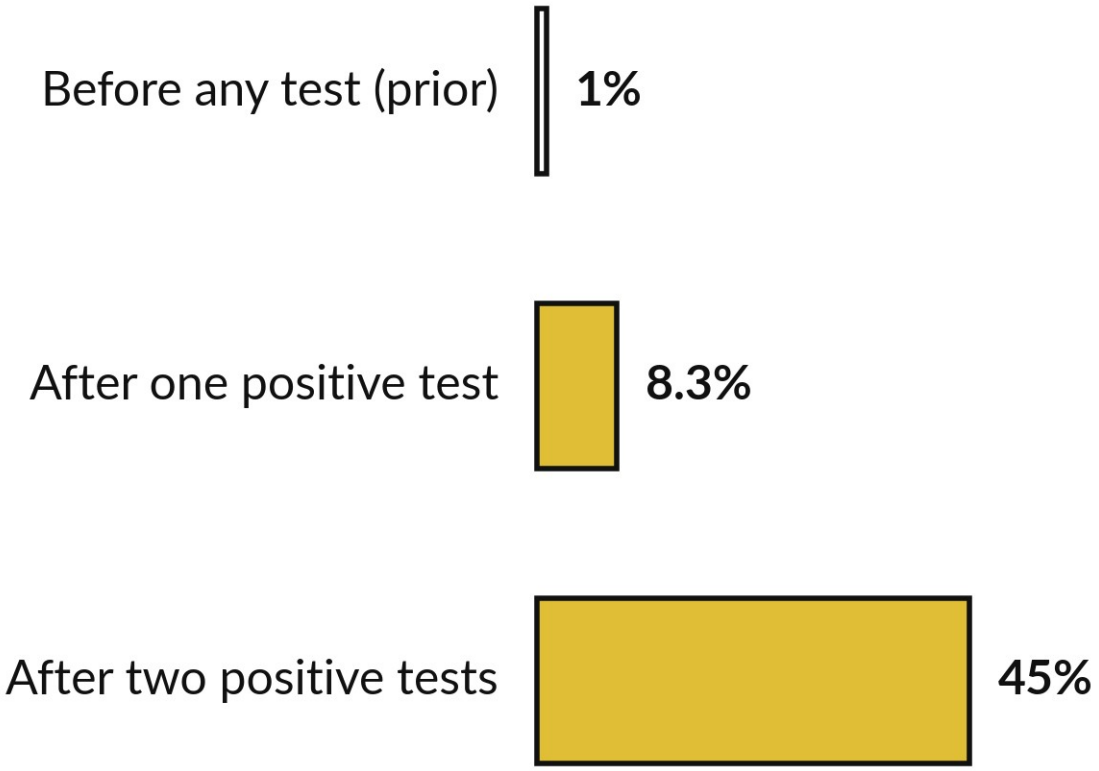
So the word carries information about spam. A filter can use it.

Natural frequencies: of 1,000 people, 9 sick people test positive and 99 healthy people test positive



Illustrative: 1% sick. Sensitivity 90% (sick found). False positive rate 10% (healthy flagged).

A positive test moves the belief from 1% to 8.3%, and a second positive test to 45%



After one positive test: 9 sick of 108 positives, 0.0833. Gold: after the evidence.

Most positives come from the many healthy people.

Two positives: 45%, if the tests err independently given sick or healthy.

Evidence updates a belief.

Bayes' theorem: prior, likelihood, evidence and posterior

Bayes' theorem

$$P(H \mid E) = \frac{P(E \mid H)\,P(H)}{P(E)}$$

Evidence

$$P(E) = P(E \mid H)\,P(H) + P(E \mid \text{not } H)\,P(\text{not } H)$$

Symbol	What it means	In our data
P(H)	Prior: belief in the hypothesis H (sick) before the evidence E (a positive test).	0.01
P(E | H)	Likelihood (here the sensitivity): how likely the evidence is if H is true.	0.90
P(E)	Marginal likelihood of the evidence, also just called the evidence: 0.9 × 0.01 + 0.1 × 0.99.	0.108
P(H | E)	Posterior: the updated belief after the evidence.	0.009 / 0.108 = 0.0833

The same 9 / 108 as the natural frequencies, with probabilities instead of counts.

The most common mistake is to swap P(E | H) and P(H | E)



P(positive | sick) = 0.90

How well the test finds sick people

The likelihood, set by the test maker



P(sick | positive) = 0.083

What you want to know after a positive result

The posterior, which also needs the prior

Ignoring the prior (the base rate) is called base rate neglect.

A tiny table of word counts is all a Naive Bayes spam filter needs

	A	B	C
	Word	**Spam emails (of 40)**	**Ham emails (of 60)**
1	free	28	6
2	prize	20	0
3	meeting	4	30

Illustrative training counts: how many emails of each class contain each word.

Prior: P(spam) = 40 / 100 = 0.4 and P(ham) = 0.6.

"free" and "prize" point to spam. "meeting" points to ham.

The gold zero is a problem: no ham email contained "prize".

Add-one smoothing turns each count into a likelihood and removes the zero

	A	B	C
	P(word | class)	Spam	Ham
1	free	0.690	0.113
2	prize	0.500	0.016
3	meeting	0.119	0.500

Add-one (Laplace) smoothing: likelihood = (count plus 1) / (emails in the class plus 2).

+ 2: present and absent each get one extra count.

prize, ham: (0 + 1) / (60 + 2) = 0.016.

A zero would make it impossible for any email with "prize" to be ham.

Conditional independence: within spam, 'free' and 'prize' are independent, though overall they are not

40 spam emails	Has "prize"	No "prize"	Total
Has "free"	14	14	28
No "free"	6	6	12
Total	20	20	40

Illustrative split of the 40 spam emails by both words.

Spam only: 14/40 = 0.70 × 0.50.

All 100: 0.14, not 0.34 × 0.20.

Independent within the class only.

Naive Bayes: multiply the prior by one likelihood per word

Naive Bayes score

$$\text{score}(c) = P(c) \prod_{w} q(w, c)$$

Posterior

$$P(\text{spam} \mid \text{words}) = \frac{\text{score}(\text{spam})}{\text{score}(\text{spam}) + \text{score}(\text{ham})}$$

Symbol	What it means	In our data
P(c)	The prior of the class c (spam or ham).	0.4 and 0.6
q(w, c)	The factor for word w: the smoothed likelihood P(w | c) if w is in the email, 1 - P(w | c) if not.	one per word
∏ over w	Capital pi: multiply over all the words, as Σ adds.	three words

Naive: within each class the words are treated as independent (conditional independence).

By hand, email 1 has 'free' and 'prize' but no 'meeting': spam with probability 0.9955

	A	B	C
	Step	**Spam**	**Ham**
1	Prior	0.400	0.600
2	"free" present	0.690	0.113
3	"prize" present	0.500	0.016
4	"meeting" absent	0.881	0.500
5	Score (product)	0.12166	0.00055
6	Posterior	0.9955	0.0045

Multiply down each column, then divide the spam score by the total.

Spam score 0.12166. Ham score 0.00055.

Posterior = 0.12166 / 0.12220 = 0.9955.

Strong evidence: spam by a wide margin.

Email 2 has 'free' and 'meeting' but no 'prize': the filter leans to ham (33% spam)

	A	B	C
	Step	**Spam**	**Ham**
1	Prior	0.400	0.600
2	"free" present	0.690	0.113
3	"meeting" present	0.119	0.500
4	"prize" absent	0.500	0.984
5	Score (product)	0.01644	0.03332
6	Posterior	0.3304	0.6696

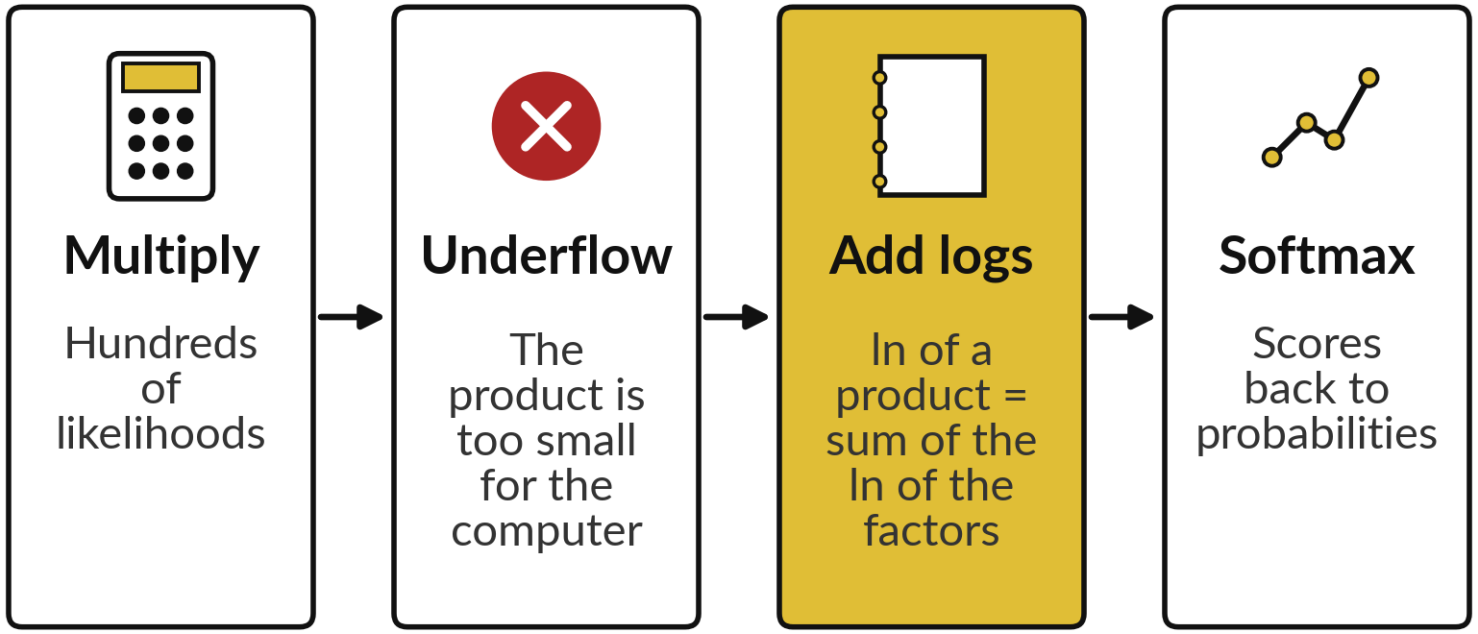
Same method. The word 'meeting' and the missing 'prize' now point to ham.

Spam score 0.01644. Ham score 0.03332.

Posterior = 0.01644 / 0.04976 = 0.3304.

Not sure: about one chance in three of spam.

Many words multiply many small numbers, so we add logarithms instead



Logarithms (Session 1) and the stable softmax (Session 12) fix it.

ln(score(spam)) = −2.107.
ln(score(ham)) = −7.512.

Softmax of the two: 0.9955, the same posterior.

Two classes: softmax = sigmoid (Session 11) of the gap, 1 / (1 + e^(ln ham - ln spam)).

Bridge: Naive Bayes from counts in NumPy

```
spam = np.array([28, 20, 4])
ham = np.array([6, 0, 30])
ls = (spam + 1) / 42
lh = (ham + 1) / 62
has = np.array([1, 1, 0])
ps = 0.4 * np.prod(np.where(has == 1, ls, 1 - ls))
ph = 0.6 * np.prod(np.where(has == 1, lh, 1 - lh))
ps / (ps + ph)
```

In plain words

Lines 1 and 2: the word counts for free, prize and meeting.

Lines 3 and 4: smoothed likelihoods.

Line 5: 1 if the word is in the email, 0 if not.

Lines 6 and 7: where picks the likelihood if present, one minus it if absent. prod multiplies.

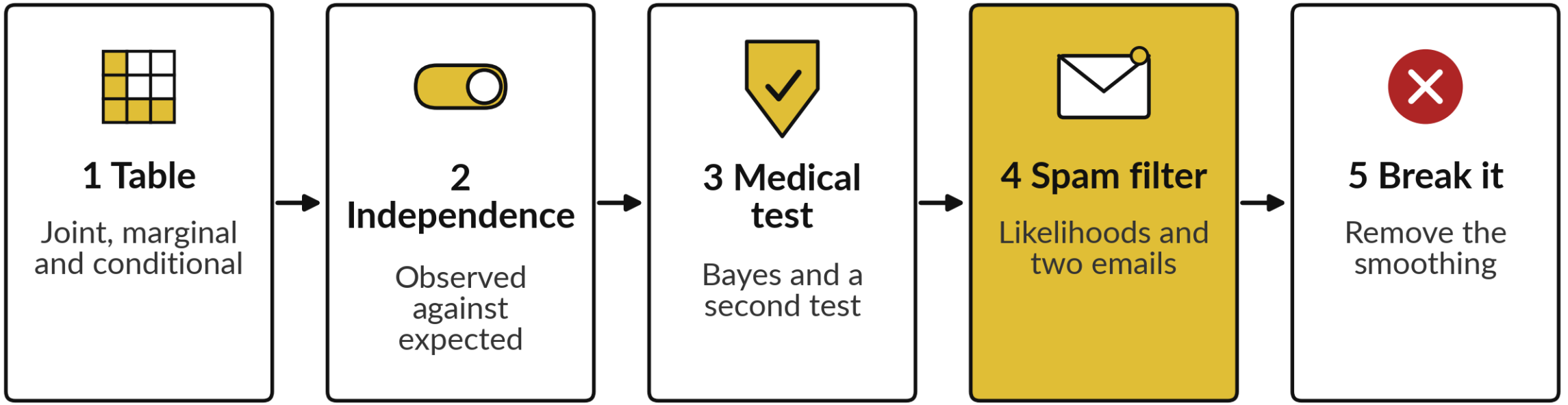
Line 8: the posterior of spam.

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Part 2: the lab

Close the slides. Open the notebook or the workbook.

The lab has five stages, in Python and in the spreadsheet



Python notebook: Session 15. Workbook sheets: Counts, Medical and NaiveBayes. Predict first, run, read, then change one thing.

Check your numbers against these before you move on

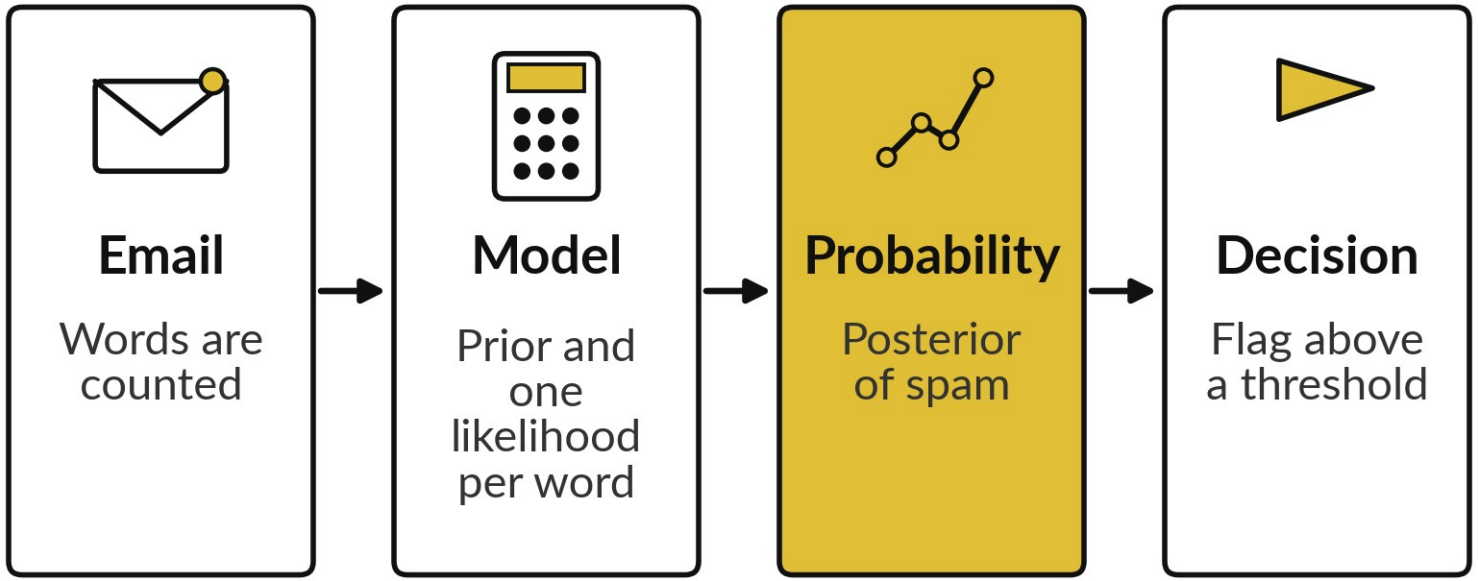
What you compute	You should see
P(spam and free), P(free), P(spam | free)	0.28, 0.34, 0.8235
P(free | spam)	0.70
Expected spam emails with 'free' if independent	13.6
Medical test: P(E), posterior after one positive test	0.108, 0.0833
Posterior after a second independent positive test	0.45
Smoothed likelihood of 'free' in spam and of 'prize' in ham	0.6905, 0.0161
Posterior spam probability of email 1 and email 2	0.9955, 0.3304
CHANGE task: email 1 with no smoothing	1.0000 (overconfident)

If your number differs, stop and find the step that went wrong. Do not change a number to match.

Errors you will meet, what they mean and what to do

	Where	What you see	What to do
✗	**Both**	P(spam | free) is 0.7	You divided by 40 instead of 34. Divide by the emails with the condition.
✗	**Both**	Posterior 0.0900 instead of 0.0833	You used 0.1 as P(E). Add both routes: 0.9 × 0.01 + 0.1 × 0.99 = 0.108. (Forgetting the false positives gives 1.0.)
✗	**Python**	Email 1 comes out as ham	The two choices in np.where are swapped. Present (has == 1) takes the likelihood, absent takes 1 minus it.
✗	**Spreadsheet**	Posterior above 1	The ham score was not added to the denominator.
✗	**Both**	Likelihood of an absent word is wrong	Use 1 - P(word | class) for a word that is not in the email.

Where this appears in AI: spam filters and medical-risk classifiers



A prior, likelihoods and a posterior run in many classifiers.

Sahami and colleagues published Bayesian junk email filtering in 1998.

A medical-risk classifier needs the base rate too.

Chat assistants are language models, not Naive Bayes.

Where this appears in AI: a stated probability should be checked against how often it is right

	A	B	C	D
	Stated confidence	Cases	Actually right	Rate
1	90%	200	150	75%
2	70%	300	216	72%
3	50%	100	49	49%

Illustrative (made-up) check: the model says 90% but is right 75% of the time.

Calibrated: stated probability equals the hit rate.

Guo and colleagues, 2017: many neural networks are over-confident.

Where this appears in AI: a fluent answer is not a probability that it is true

What the model computes

The probability of the next token given the text

High for a likely wording

What it does not give you

The probability that the statement is true

A warning that a fluent sentence is false

Survey: Ji and colleagues, 2023

Check important claims against a source or your own calculation.

Where this appears in AI: check an assistant on the medical test

8 minutes **Check an AI assistant**

Ask an AI assistant the medical test question with our made-up numbers.

Say: 1% are sick, 90% of sick people test positive, and 10% of healthy people do too.

Ask: I tested positive. How likely am I sick?

Compare with 9 / 108 = 0.0833.

Ask how sure it is.

You keep

A note: its answer, the correct 8.3%, and what its confidence meant.

Exit check: three questions

Question	Short answer
In the 100 emails, P(free | spam) = 0.70 and P(spam | free) = 0.82. Why do they differ?	They condition on different groups: 40 spam emails against 34 emails with the word.
A disease affects 1 in 1,000. A test finds 99% of sick people and wrongly flags 1% of healthy people. Roughly how likely is a positive person to be sick?	About 9%. Of 1,000 people, about 1 sick positive and about 10 healthy positives: 1 / 11.
A language model gives a fluent answer. Is that the probability that it is true?	No. It reflects how likely the wording is. Check the claim.

Save the lab and keep your key numbers

What you keep from today

Keep your saved lab and these numbers: P(spam given free) = 0.8235, the medical test posterior 0.0833, and the spam posteriors 0.9955 and 0.3304.

Our question today

A test finds 90% of sick people and wrongly flags 10% of healthy people. You test positive. How likely are you sick?

Next session

Session 16: maximum likelihood and MAP. We see why squared error and cross-entropy are the right losses.

Take away

Saved lab
Notebook or workbook

Key numbers
See the check table

Next session
Maximum likelihood

Bring
A question on Bayes